# Definite Integration: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

## 1. Meaning and basic rules

**Formulas**

$\int_a^b f(x)\,dx = F(b) - F(a)$, where $F' = f$ and $F$ is continuous on $[a, b]$
$\int_a^b f(x)dx = \int_a^b f(t)dt$, $\int_a^b f = -\int_b^a f$, $\int_a^a f = 0$

KEY POINTS

- Value = algebraic sum of areas: above the $x$-axis $+$, below $-$
- $f \geq 0$ on $[a, b] \Rightarrow \int_a^b f \geq 0$; $f$ continuous and $\int_a^b f = 0 \Rightarrow f$ has a root in $(a, b)$
- Limit inside the integral: allowed when the limits do not depend on $n$
- Parts in definite form: $\int_a^b uv'\,dx = \big[uv\big]_a^b - \int_a^b u'v\,dx$

**Watch out**

Antiderivative must not break inside $[a, b]$ ($\tan x$ at $\frac{\pi}{2}$, $\tan\frac{x}{2}$ at $\pi$, $\tan^{-1}\frac{1}{x}$ at 0)

## 2. Substitution in definite integrals

KEY POINTS

- Change the limits with the substitution; do not return to $x$
- Substitution must be continuous and one-one on the whole interval
- Limits belong to the variable in the differential: $\int_{\ln x=0}^{1} x\,d(\ln x) = \int_1^e dx$
- Reciprocal limits ($\frac{1}{a}$ to $a$, or 0 to $\infty$): put $x = \frac{1}{t}$ and add
- Equal new limits $\Rightarrow$ integral $= 0$

**Formulas**

Inverse functions: $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(y)dy = b\,f(b) - a\,f(a)$

## 3. Values to remember

**Formulas**

$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = 1,\ \frac{\pi}{4},\ \frac{2}{3},\ \frac{3\pi}{16}$ $(n = 1, 2, 3, 4)$
$\int_0^{\pi/2} \ln\sin x\,dx = \int_0^{\pi/2} \ln\cos x\,dx = -\frac{\pi}{2}\ln 2$, $\int_0^{\pi/4} \ln(1 + \tan x)dx = \frac{\pi}{8}\ln 2$

## 4. P3: splitting the interval

**Formulas**

$\int_a^b f = \int_a^c f + \int_c^b f$ ($c$ anywhere)

STEPS

1. Find the break points: zeros inside $|\ |$, integer values inside $[\ ]$ or $\{\ \}$, points where max/min or the formula changes
2. Write the function without $|\ |$, $[\ ]$ on each piece
3. Integrate piece by piece and add

KEY POINTS

- $\int_0^n [x]dx = \frac{n(n-1)}{2}$, $\int_0^n \{x\}dx = \frac{n}{2}$ $(n \in \mathbb{N})$
- $f(y) + f(y + k) = c$: pieces of length $k$, each pair adds to $ck$

## 5. P4: odd and even functions

**Formulas**

$\int_{-a}^{a} f = \int_0^a [f(x) + f(-x)]dx$; $= 2\int_0^a f$ (even), $= 0$ (odd)

KEY POINTS

- Use when limits are $-a$ to $a$; limits $c \pm a$: put $x - c = t$ first
- Split into even part + odd part; odd: $x^{2k+1}$, $\sin x$, $\tan x$, $\ln \frac{1-x}{1+x}$, $\ln(x + \sqrt{x^2+1})$
- $\int_{-a}^{a} \frac{f(x)}{1+b^x}dx = \int_0^a f(x)dx$ for even $f$ (also $\frac{1}{1+e^{g(x)}}$ with $g$ odd)
- $\int_0^x f(t)dt$ is even if $f$ is odd, and odd if $f$ is even

## 6. P5: King property

**Formulas**

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx = \frac{b-a}{2}$; $f(a+b-x) = f(x) \Rightarrow \int_a^b xf(x)dx = \frac{a+b}{2}\int_a^b f(x)dx$

STEPS

1. Replace $x$ by $a+b-x$ (limits unchanged)
2. Add (or subtract) the two forms of $I$
3. Evaluate the simpler integral; divide by 2

KEY POINTS

- Use when the denominator is unchanged by king, for $x \cdot f(\sin x)$ on $[0, \pi]$, and for $\ln(1+\tan x)$-type
- $f(x) + f(a+b-x) = c \Rightarrow \int_a^b f = \frac{c(b-a)}{2}$

## 7. P6: Queen property

**Formulas**

$\int_0^{2a} f = \int_0^a [f(x) + f(2a-x)]dx$; $= 2\int_0^a f$ if $f(2a-x) = f(x)$, $= 0$ if $f(2a-x) = -f(x)$

KEY POINTS

- Lower limit must be 0; it halves the upper limit
- Typical: sin, cos powers on $[0, \pi]$, $[0, 2\pi]$; repeat until $[0, \frac{\pi}{2}]$, then Wallis or king

## 8. P7: periodic functions

**Formulas**

$\int_0^{nT} f = n\int_0^T f$, $\int_a^{a+nT} f = n\int_0^T f$, $\int_{nT}^{a+nT} f = \int_0^a f$

KEY POINTS

- Periods: $|\sin x|$, $|\cos x|$, $\sin^2 x$, $\tan x$: $\pi$; $|\sin x| + |\cos x|$: $\frac{\pi}{2}$; $\{x\}$, $x - [x]$: 1
- $f(x) + f(x+k) = c \Rightarrow f$ has period $2k$
- Remove whole periods, then integrate the remaining part

## 9. Leibnitz rule

**Formulas**

$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)dt = f(h(x))h'(x) - f(g(x))g'(x)$; $\frac{d}{dx}\int f(x)dx = f(x)$

KEY POINTS

- Integrand must be free of $x$: take $x$ out or substitute first
- $\frac{0}{0}$ limits with integrals: L'Hospital + Leibnitz
- $F(x) = \int_a^x f$: $F' = f$ gives monotonicity, maxima, minima, range of $F$

## 10. Determination of function

STEPS

1. Constant limits: put $\int_a^b f(t)dt = k$ (or $A$, $B$ for several parts), substitute back, solve for $k$
2. Variable limits: differentiate (Leibnitz) to get a differential equation
3. Initial value: put $x$ equal to the lower limit

KEY POINTS

- $f(x)$, $f(\frac{1}{x})$ given together: replace $x$ by $\frac{1}{x}$ and solve the two equations
- $f'(x) = f'(a-x) \Rightarrow f(x) + f(a-x) = \text{constant}$

## 11. Limit of a sum

**Formulas**

$$\lim_{n\to\infty} \frac{1}{n} \sum_r f\left(\frac{r}{n}\right) = \int_\alpha^\beta f(x)\,dx,\ \alpha = \lim \tfrac{r_{\text{first}}}{n},\ \beta = \lim \tfrac{r_{\text{last}}}{n}$$

STEPS

1. Write the $r$th term; take out $\frac{1}{n}$ (highest power of $n$)
2. $\frac{r}{n} \to x$, $\frac{1}{n} \to dx$, $\lim \sum \to \int$; find the limits
3. Continued product: take ln first

KEY POINTS

- $f$ increasing: left sum $< \int_0^1 f <$ right sum; decreasing: reverse

## 12. Estimation

**Formulas**

$m \le f \le M$ on $[a,b] \Rightarrow m(b-a) \le \int_a^b f \le M(b-a)$

Concave up: $(b-a) f\left(\frac{a+b}{2}\right) < \int_a^b f < (b-a)\frac{f(a)+f(b)}{2}$

$f_1 \le f \le f_2 \Rightarrow \int f_1 \le \int f \le \int f_2$; $\left|\int_a^b f\right| \le \int_a^b |f|$

KEY POINTS

- Monotonic: use end values; otherwise find max/min by derivative
- Split the interval for sharper bounds; midpoint rule $\frac{b-a}{4}\left[f(a) + 2f(\frac{a+b}{2}) + f(b)\right]$
- Useful comparisons: on $(0,1)$ $x^n$ decreases with $n$; $\sin x < x$; $1 + x < e^x$

## 13. Wallis and reduction

**Formulas**

$$\int_0^{\pi/2} \sin^n x \cos^m x\,dx = \frac{[(n-1)(n-3)\cdots][(m-1)(m-3)\cdots]}{(m+n)(m+n-2)\cdots} \cdot K,\ K = \frac{\pi}{2}\ (m,\ n \text{ both even}),\ \text{else } 1$$

$$I_n = \int_0^{\pi/2} \sin^n x\,dx = \frac{n-1}{n} I_{n-2};\ \int_0^{\pi/4} \tan^n x:\ I_n + I_{n-2} = \frac{1}{n-1};\ \int_1^e (\ln x)^n:\ I_n = e - nI_{n-1}$$

KEY POINTS

- Reduction: parts with 1 as second function or one factor split off; boundary term often 0
- Other limits: use queen / king / periodicity to reach $[0, \frac{\pi}{2}]$ first

## 14. Common mistakes checklist

STEPS

1. Did the antiderivative or substitution break inside the interval? Split there.
2. Did you change the limits with the substitution?
3. $|f|$, $[f]$: found all break points inside the interval?
4. King: keep the same limits; queen: lower limit 0
5. Leibnitz: integrand free of $x$? both limit terms with chain factor?
6. Limit of sum: correct limits from first and last $r$
7. Wallis: $K = \frac{\pi}{2}$ only when both powers are even